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Studies on $ZnBr_2$-mediated synthesis of axially chiral aryl-substituted allenes from terminal alkynes, aromatic aldehydes and (*S*)-α,α-diphenylprolinol

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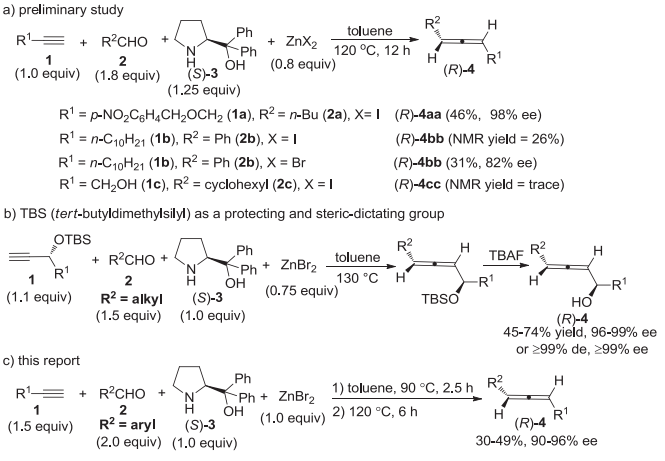
ABSTRACT

The studies on the $ZnBr_2$-mediated synthesis of axially chiral aryl-substituted allenes from terminal alkynes, aromatic aldehydes, and commercially available chiral amine (*S*)-α,α-diphenylprolinol were conducted. Axially chiral aryl-substituted allenes can be obtained in moderate yields with up to 96% ee.

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1. Introduction

Allenes have become a very important class of basic organic compounds in organic synthesis, pharmaceutical chemistry, and materials science.[1] However, efficient methods for the synthesis of optically active allenes, especially those with an axial chirality, are still highly desirable.[2–4] Recently, we have developed two convenient approaches for the synthesis of axially chiral 1,3-disubstituted allenes from terminal alkynes and aldehydes utilizing the 'chiral ligand' approach[4a] or 'chiral amine' approach[4b,c] based on our previously reported ZnI_2-promoted one-pot synthesis of 1,3-disubstituted allenes from terminal alkynes, aldehydes and morphine.[5] While the 'chiral amine' approach utilizing (*S*)-α,α-diphenylprolinol **3** represents a highly efficient protocol for the synthesis of axially chiral 1,3-disubstituted allenes from terminal alkynes and aldehydes, our preliminary study revealed that the substrate scope is quite limited. For example, the reaction of aromatic aldehydes such as benzaldehyde **2b** afforded the corresponding product (*R*)-**4bb** in very low yield with moderate ee whereas the reaction of unprotected propargyl alcohol **1c** only afforded trace amount of the product as determined by ^1H NMR analysis (a, Scheme 1). To expand the scope as well as demonstrate the practicability of this 'chiral amine' approach, we recently developed a highly enantioselective synthesis of axially chiral α-allenols from TBS-protected propargylic alcohols and aldehydes in the presence of (*S*)-α,α-diphenylprolinol **3** or its enantiomer, in which the



Scheme 1. Synthesis of axially chiral allenes using (*S*)-α,α-diphenylprolinol **3**.

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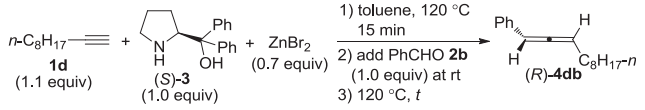
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TBS group is not only acting as a protecting group, but also as a steric-dictating group.[4b] Again, aliphatic aldehydes proved to be the more compatible with this procedure as the reactions of aromatic aldehydes provided very low yields of the corresponding aryl-substituted allenes (b, Scheme 1).[4b]

2. Results and discussion

Surprisingly, we noticed that Periasamy et al. reported a similar procedure for the synthesis of axially chiral 1,3-disubstituted allenes from both aliphatic and aromatic aldehydes in the presence of (*S*)-α,α-diphenylprolinol **3**.[6] In their paper, they claimed that stirring the solution of $ZnBr_2$, (*S*)-α,α-diphenylprolinol **3**, and terminal alkynes in toluene for 10 min at 120 °C followed by the addition of aldehydes at room temperature and then gradually heated to 120 °C from room temperature are very important for the high yields and ees for aryl-substituted allenes. However, we have not been able to reproduce their results in terms of yield and ee, for example, in the reaction of 1-decyne **1d** with benzaldehyde **2b**, even after examining with several different well-known sources of $ZnBr_2$ with different purity (entry 1 vs entries 2–5, Table S1 in Supplementary data). Furthermore, we also noticed that the reaction time had an obvious influence on both the yield and enantiomeric excess of the aryl-substituted allene product (*R*)-**4db** (Table 1): when the reaction time was prolonged from 4 h to 16 h, the yield of (*R*)-**4db** improved from less than 7–33% whereas the ee of (*R*)-**4db** dropped from 96% to 84%, indicating that the axially chiral aryl-substituted allene product (*R*)-**4db** underwent in situ racemization under this set of reaction conditions.[7] Herein we wish to disclose our own observation on this $ZnBr_2$-mediated reaction.

Table 1
The influence of reaction time on the reaction of 1-decyne **1d**, benzaldehyde **2b**, and (*S*)-α,α-diphenylprolinol **3**[a]

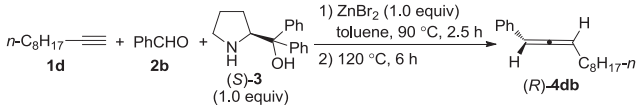


Entry	*t* (h)	(*R*)-**4db**	
		Yield[b] (%)	ee[c] (%)
1	4	7[d]	96
2	6	13	94
3	8	25	94
4[e]	10	32	91
5	16	33	84

[a] The reactions were carried out with $ZnBr_2$ (0.7 mmol), (*S*)-**3** (1.0 mmol), 1-decyne **1d** (1.1 mmol), benzaldehyde **2b** (1 mmol), and toluene (3 mL).
[b] Isolated yield of (*R*)-**4db**.
[c] Determined by chiral HPLC analysis.
[d] Contaminated with unreacted **1d**, the molar ratio of (*R*)-**4db** to **1d** is 1:3.17 as determined by ^1H NMR.
[e] Heated for 30 min at 120 °C before adding benzaldehyde **2b**.

After extensive optimization of the reaction conditions, we eventually found that when the reaction mixture of 1.3 equiv of 1-decyne **1d**, 1.0 equiv of benzaldehyde **2b**, (*S*)-α,α-diphenylprolinol **3**, and $ZnBr_2$ was heated for 2.5 h at 90 °C followed by 6 h at 120 °C, the yield of (*R*)-**4db** could be improved to 43% and the ee could be maintained at 91% (entry 1, Table 2). The ratio of 1-decyne **1d** to benzaldehyde **2b** was also found to be crucial for the yield as well as the ee of (*R*)-**4db**. Increasing the amount of 1-decyne **1d** was beneficial for the yield of the reaction (entry 1 vs 2, 5 vs 6, Table 2) whereas increasing the amount of benzaldehyde **2b** improved the enantioselectivity but diminished the yield of (*R*)-**4db** (entries 2–4, Table 2). In view of both the yield and enantioselectivity of the reaction, 1.5 equiv of terminal alkyne and 2.0 equiv of aromatic aldehydes in the presence of 1.0 equiv of (*S*)-α,α-diphenylprolinol **3** and $ZnBr_2$ were defined as the standard conditions for substrates scope evaluation (entry 3, Table 2).

Table 2
Optimization of the reaction conditions[a]

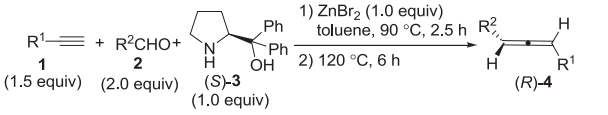


Entry	Ratio of **1d**/**2b**/(*S*)-**3**	(*R*)-**4db**	
		Yield[b] (%)	ee[c] (%)
1	1.3:1:1	43	91
2	1.5:1:1	47	91
3	*1.5:2:1*	*42*	*94*
4	1.5:2.2:1	33	96
5	1.2:2:1	39	94
6	1.7:2:1	44	94

[a] The reactions were carried out by adding $ZnBr_2$ (1.0 mmol), (*S*)-**3** (1.0 mmol), 1-decyne **1d** (1.2–1.7 mmol)/toluene (2 mL), and benzaldehyde **2b** (1–2.2 mmol)/toluene (1 mL) sequentially under Ar atmosphere at rt, heating for 2.5 h at 90 °C followed by 6 h at 120 °C.
[b] Isolated yield of (*R*)-**4db**.
[c] Determined by chiral HPLC analysis.

The optimized reaction conditions are applicable to a reasonably range of electron-poor and -rich aromatic aldehydes (Table 3): halogen-substituted aromatic aldehydes with a Br, Cl or F atom at *meta*- or *para*-position are well tolerated, affording the corresponding axially chiral aryl-substituted allenes in moderate yields and up to 96% ee (entries 2–6, Table 3). Aromatic aldehydes containing a CH_3, CF_3 or MeO group can also be smoothly transformed into the desired products, furnishing the axially chiral allenes in practical yields and with 92–95% ee (entries 7–10, Table 3). In addition to normal alkyl-substituted terminal alkynes **1b** and **1d**, cycloalkyl-substituted terminal alkyne **1e** may also be employed to yield (*R*)-**4eb** in 44% yield with 95% ee (entry 12, Table 3). It should be noted that aliphatic aldehyde **2m** is also compatible with the

Table 3
Synthesis of axially chiral allenes from 1-alkynes **1**, aldehydes **2** and (*S*)-**3**[a]



Entry	R^1	R^2	(*R*)-**4**	
			Yield[b] (%)	ee[c] (%)
1	*n*-C_8H_{17} (**1d**)	Ph (**2b**)	42 (*R*-**4db**)	94
2	*n*-C_8H_{17} (**1d**)	*p*-BrC_6H_4 (**2d**)	30 (*R*-**4dd**)	96
3	*n*-C_8H_{17} (**1d**)	*m*-BrC_6H_4 (**2e**)	36 (*R*-**4de**)	95
4	*n*-C_8H_{17} (**1d**)	*p*-ClC_6H_4 (**2f**)	48 (*R*-**4df**)	91
5	*n*-C_8H_{17} (**1d**)	*m*-ClC_6H_4 (**2g**)	39 (*R*-**4dg**)	90
6	*n*-C_8H_{17} (**1d**)	*p*-FC_6H_4 (**2h**)	47 (*R*-**4dh**)	92
7	*n*-C_8H_{17} (**1d**)	*p*-$CH_3C_6H_4$ (**2i**)	32 (*R*-**4di**)	94
8	*n*-C_8H_{17} (**1d**)	*m*-$CH_3C_6H_4$ (**2j**)	44 (*R*-**4dj**)	92
9	*n*-C_8H_{17} (**1d**)	*p*-$CF_3C_6H_4$ (**2k**)	31 (*R*-**4dk**)	95
10	*n*-C_8H_{17} (**1d**)	*m*-$MeOC_6H_4$ (**2l**)	49 (*R*-**4dl**)	94
11	*n*-$C_{10}H_{21}$ (**1b**)	Ph (**2b**)	46 (*R*-**4bb**)	93
12	*c*-C_6H_{11} (**1e**)	Ph (**2b**)	44 (*R*-**4eb**)	95
13	CH_2OBn (**1f**)	*c*-C_6H_{11} (**2m**)	58 (*R*-**4fm**)	98

[a] The reactions were carried out by adding $ZnBr_2$ (1.0 mmol), (*S*)-**3** (1.0 mmol), 1-alkynes **1** (1.5 mmol)/toluene (2 mL), and aldehyde **2** (2.0 mmol)/toluene (1 mL) sequentially under Ar atmosphere at rt, heating for 2.5 h at 90 °C followed by 6 h at 120 °C.
[b] Isolated yield of (*R*)-**4**.
[c] Determined by chiral HPLC analysis.

current procedure, affording (*R*)-**4fm** in 58% yield with slightly higher ee than those of aromatic aldehydes (entry 13, Table 3).

3. Conclusion

We have studied the $ZnBr_2$-mediated synthesis of axially chiral aryl-substituted allenes from terminal alkynes, aromatic aldehydes, and commercially available chiral amine (*S*)-α,α-diphenylprolinol **3** in detail. Due to the easy reproducibility of the current procedure, ready availability of the starting materials, as well as the potentials of the axially chiral allenes products,[1,8] this protocol will be of interest for the synthetic community. Further studies in this area are ongoing in our laboratory.

4. Experimental section

4.1. General remarks

NMR spectra were recorded with a Bruker AM-300 or AM-400 instrument. Chemical shifts are reported in relative to TMS [^{1}H: δ=0.00 ppm (TMS)], solvent residual peaks [^{13}C: δ=77.0 ppm ($CDCl_3$)] or internal standard [^{19}F: δ=0.00 ppm ($CFCl_3$)]. IR spectra were measured with a Bruker Tensor 27 instrument. MS analyses were performed with a Agilent Technologies 5975C instrument. HRMS analyses were performed with a Waters GCT Premier instrument. Specific optical rotations were measured with OA PolAAr 3005 instrument. HPLC analyses were performed with Waters 2489 detector and Waters 515 pump. All reactions were carried out in oven-dried Schlenk tubes. $ZnBr_2$ was purchased from Acros (98%), Alfa Aesar (98% or 99.9%) or Sinopharm, China (98%) and kept in glovebox; 1-dodecyne **1b** (97%) and 1-decyne **1d** (98%) were purchased from Alfa Aesar; (*S*)-α,α-diphenylprolinol **3** (98%) was purchased from Shanghai Darui Finechemicals. Liquid aldehydes were distilled right before use. Toluene was dried over sodium wire with benzophenone as indicator and distilled freshly before use. Other reagents were used as received from commercial sources. All the temperatures are referred to the oil baths used. The petroleum ether (30–60 °C) for chromatography was distilled before use.

4.2. Preparation of (*R*)-4db–(*R*)-4fb (Table 3)

*4.2.1. (R)-1-Phenylundeca-1,2-diene (R-**4db**). Typical procedure*: To an oven-dried Schlenk tube was added $ZnBr_2$ (229.9 mg, 1.0 mmol, 98% from Alfa) inside a glovebox, the Schlenk tube was then taken out of the glovebox and heated the $ZnBr_2$ until melting under vacuum with a heating gun. (*S*)-**3** (258.4 mg, 1.0 mmol, 98%), **1d** (212.1 mg, 1.5 mmol, 98%)/toluene (2 mL), and **2b** (212.0 mg, 2.0 mmol)/toluene (1 mL) were then added sequentially under argon atmosphere. The Schlenk tube was then equipped with a condenser and placed in a pre-heated oil bath at 90 °C with stirring for 2.5 h. The reaction mixture was then heated to 120 °C in about 10 min and further stirred for 6 h at 120 °C. After cooling to room temperature, the crude reaction mixture was filtered through a short pad of silica gel and eluented with ether (20 mL). After evaporation, the residue was purified by chromatography on silica gel to afford (*R*)-**4db**[6] (95.8 mg, 42%) (eluant: petroleum ether) as a colorless liquid: 94% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 0.7 mL/min, λ=214 nm, t_R (major)=8.8 min, t_R (minor)=9.6 min); $[\alpha]_D^{20}$ −227.5 (*c* 0.99, $CHCl_3$) (reported value for 98% ee: $[\alpha]_D^{25}$ −225.1 (*c* 0.50, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.34–7.24 (m, 4H, Ar–H), 7.24–7.12 (m, 1H, Ar–H), 6.11 (dt, J_1=6.0 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.55 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.12 (qd, J_1=7.1 Hz, J_2=2.9 Hz, 2H, CH_2), 1.55–1.14 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.5 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=205.1, 135.2, 128.5, 126.6, 95.1, 94.5, 31.9, 29.4, 29.3, 29.20, 29.17, 28.8, 22.7, 14.1; MS (EI) *m*/*z* (%): 228.2 (M^+, 1.00), 157 ($M^+-C_5H_{11}$, 1.79), 130.1 (100); IR (neat): ν=2924, 2854, 1950, 1599, 1496, 1459, 1261, 1072, 1028 cm^{-1}; HRMS calcd for $C_{17}H_{24}$ [M^+]: 228.1878, found: 228.1881.

The following compounds were prepared according to the above *typical procedure*. The corresponding racemic compounds were prepared according to our early report.[5]

*4.2.2. (R)-1-(4-Bromophenyl)undeca-1,2-diene (R-**4dd**).* The reaction of $ZnBr_2$ (229.3 mg, 1.0 mmol, 98% from Alfa), **1d** (214.1 mg, 1.5 mmol, 98%), **2d** (377.9 mg, 2.0 mmol, 98%), (*S*)-**3** (258.3 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4dd**[6] (91.0 mg, 30%) (eluant: petroleum ether) as a colorless liquid: 96% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=400:1, 0.5 mL/min, λ=214 nm, t_R (minor)=7.2 min, t_R (major)=8.2 min); $[\alpha]_D^{19}$ −198.1 (*c* 1.03, $CHCl_3$) (reported value for 90% ee: $[\alpha]_D^{25}$ −139.2 (*c* 0.50, $CHCl_3$)); ^{1}H NMR (400 MHz, $CDCl_3$) δ=7.40 (d, *J*=8.4 Hz, 2H, Ar–H), 7.14 (d, *J*=8.0 Hz, 2H, Ar–H), 6.06 (dt, J_1=6.4 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.56 (q, *J*=6.5 Hz, 1H, one proton from CH=C=CH), 2.12 (qd, J_1=7.1 Hz, J_2=2.7 Hz, 2H, CH_2), 1.52–1.40 (m, 2H, CH_2), 1.40–1.12 (m, 10H, $CH_2\times5$), 0.87 (t, *J*=6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ=205.2, 134.2, 131.6, 128.1, 120.1, 95.6, 93.7, 31.8, 29.4, 29.3, 29.2, 29.1, 28.6, 22.7, 14.1; MS (EI) *m*/*z* (%): 308 (M^+(^{81}Br), 1.95), 306 (M^+(^{79}Br), 1.68), 129 (100); IR (neat): ν=2924, 2853, 1950, 1488, 1465, 1386, 1070, 1010 cm^{-1}; HRMS calcd for $C_{17}H_{23}{}^{79}Br$ [M^+]: 306.0983, found: 306.0985.

*4.2.3. (R)-1-(3-Bromophenyl)undeca-1,2-diene (R-**4de**).* The reaction of $ZnBr_2$ (229.8 mg, 1.0 mmol, 98% from Alfa), **1d** (211.1 mg, 1.5 mmol, 98%), **2e** (370.1 mg, 2.0 mmol), (*S*)-**3** (258.2 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4de** (109.5 mg, 36%) (eluant: petroleum ether) as a colorless liquid: 95% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100/0, 0.5 mL/min, λ=214 nm, t_R (major)=13.0 min, t_R (minor)=15.0 min); $[\alpha]_D^{20}$ −192.9 (*c* 1.01, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ=7.43 (s, 1H, Ar–H), 7.29 (dt, J_1=7.6 Hz, J_2=1.5 Hz, 1H, Ar–H), 7.21–7.12 (m, 2H, Ar–H), 6.04 (dt, J_1=6.4 Hz, J_2=3.1 Hz, 1H, one proton from CH=C=CH), 5.60 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.12 (qd, J_1=7.1 Hz, J_2=3.0 Hz, 2H, CH_2), 1.53–1.40 (m, 2H, CH_2), 1.40–1.15 (m, 10H, $CH_2\times5$), 0.87 (t, *J*=6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ=205.4, 137.5, 129.9, 129.4, 129.3, 125.1, 122.7, 95.7, 93.5, 31.8, 29.4, 29.3, 29.2, 29.0, 28.6, 22.7, 14.1; MS (EI) *m*/*z* (%): 308 (M^+(^{81}Br), 1.17), 306 (M^+(^{79}Br), 1.25), 129 (100); IR (neat): ν=2955, 2924, 2853, 1950, 1590, 1566, 1475, 1069 cm^{-1}; HRMS calcd for $C_{17}H_{23}{}^{79}Br$ [M^+]: 306.0983, found: 306.0984.

*4.2.4. (R)-1-(4-Chlorophenyl)undeca-1,2-diene (R-**4df**).* The reaction of $ZnBr_2$ (229.1 mg, 1.0 mmol, 98% from Alfa), **1d** (212.2 mg, 1.5 mmol, 98%), **2f** (287.7 mg, 2.0 mmol, 98%), (*S*)-**3** (258.4 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4df**[6] (125.1 mg, 48%) (eluant: petroleum ether) as a colorless liquid: 91% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 0.7 mL/min, λ=254 nm, t_R (minor)=10.9 min, t_R (major)=14.6 min); $[\alpha]_D^{19}$ −204.1 (*c* 1.04, $CHCl_3$) (reported value for 90% ee: $[\alpha]_D^{25}$ −167.8 (*c* 0.50, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.30–7.13 (m, 4H, Ar–H), 6.07 (dt, J_1=6.3 Hz, J_2=3.1 Hz, 1H, one proton from CH=C=CH), 5.56 (q, *J*=6.6 Hz, 1H, one proton from CH=C=CH), 2.11 (qd, J_1=7.1 Hz, J_2=2.8 Hz, 2H, CH_2), 1.53–1.13 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=205.2, 133.7, 132.1, 128.6, 127.7, 95.5, 93.7, 31.9, 29.4, 29.3, 29.2, 29.1, 28.6, 22.7, 14.1; MS (EI) *m*/*z* (%): 264 (M^+(^{37}Cl), 1.47), 262 (M^+(^{35}Cl), 4.07), 129 (100); IR (neat): ν=2854, 1951, 1491, 1092, 1014 cm^{-1}; HRMS calcd for $C_{17}H_{23}{}^{35}Cl$ [M^+]: 262.1488, found: 262.1484.

*4.2.5. (R)-1-(3-Chlorophenyl)undeca-1,2-diene (R-**4dg**).* The reaction of $ZnBr_2$ (230.3 mg, 1.0 mmol, 98% from Alfa), **1d** (212.4 mg, 1.5 mmol, 98%), **2g** (288.1 mg, 2.0 mmol), (*S*)-**3** (258.7 mg, 1.0 mmol,

98%), and toluene (3 mL) afforded (*R*)-**4dg** (102.5 mg, 39%) (eluant: petroleum ether) as a colorless liquid: 90% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=400:1, 0.7 mL/min, λ=214 nm, t_R (major)=3.9 min, t_R (minor)=4.4 min); $[\alpha]_D^{20}$ −220.3 (*c* 1.06, $CHCl_3$); ^{1}H NMR (400 MHz, $CDCl_3$) δ=7.26 (s, 1H, Ar−H), 7.22−7.15 (m, 1H, Ar−H), 7.15−7.09 (m, 2H, Ar−H), 6.04 (dt, J_1=6.4 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.58 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.11 (qd, J_1=7.0 Hz, J_2=2.9 Hz, 2H, CH_2), 1.53−1.40 (m, 2H, CH_2), 1.40−1.14 (m, 10H, $CH_2\times5$), 0.87 (t, *J*=6.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ=205.4, 137.2, 134.5, 129.6, 126.5, 126.4, 124.7, 95.6, 93.7, 31.9, 29.4, 29.3, 29.2, 29.1, 28.6, 22.7, 14.1; MS (EI) *m*/*z* (%):264 (M^+(^{37}Cl), 0.66), 262 (M^+(^{35}Cl), 2.07), 129 (100); IR (neat): ν=2925, 2854, 1950, 1594, 1571, 1477, 1077 cm^{-1}; HRMS calcd for $C_{17}H_{23}{}^{35}Cl$ [M^+]: 262.1488, found: 262.1486.

*4.2.6. (R)-1-(4-Fluorophenyl)undeca-1,2-diene (R-**4dh**).* The reaction of $ZnBr_2$ (229.9 mg, 1.0 mmol, 98% from Alfa), **1d** (211.7 mg, 1.5 mmol, 98%), **2h** (249.1 mg, 2.0 mmol), (*S*)-**3** (259.3 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4dh**[6] (116.7 mg, 47%) (eluant: petroleum ether) as a colorless liquid: 92% ee (HPLC conditions: Chiralcel OJ-H column, hexane/*i*-PrOH=100:0, 0.6 mL/min, λ=214 nm, t_R (major)=8.2 min, t_R (minor)=9.4 min); $[\alpha]_D^{23}$ −180.8 (*c* 1.00, $CHCl_3$) (reported value for 90% ee: $[\alpha]_D^{25}$ −146.5 (*c* 0.50, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.27−7.17 (m, 2H, Ar−H), 6.97 (t, *J*=8.7 Hz, 2H, Ar−H), 6.08 (dt, J_1=6.3 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.55 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.11 (qd, J_1=7.0 Hz, J_2=2.9 Hz, 2H, CH_2), 1.53−1.13 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.3 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=204.9 (d, *J*=1.7 Hz), 161.7 (d, *J*=243.2 Hz), 131.1 (d, *J*=3.2 Hz), 127.9 (d, *J*=8.6 Hz), 115.4 (d, *J*=21.8 Hz), 95.3, 93.6, 31.9, 29.4, 29.3, 29.2, 29.1, 28.8, 22.7, 14.1; ^{19}F NMR (282 MHz, $CDCl_3$) δ=−116.4; MS (EI) *m*/*z* (%): 246 (M^+, 3.44), 175 ($M^+-C_5H_{11}$, 3.77), 148 (100); IR (neat): ν=2926, 2855, 1950, 1599, 1509, 1230, 1156 cm^{-1}; HRMS calcd for $C_{17}H_{23}F$ [M^+]: 246.1784, found: 246.1787.

*4.2.7. (R)-1-(4-Methylphenyl)undeca-1,2-diene (R-**4di**).* The reaction of $ZnBr_2$ (229.5 mg, 1.0 mmol, 98% from Alfa), **1d** (212.1 mg, 1.5 mmol, 98%), **2i** (240.1 mg, 2.0 mmol), (*S*)-**3** (258.3 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4di**[6] (78.6 mg, 32%) (eluant: petroleum ether) as a colorless liquid: 94% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 0.7 mL/min, λ=254 nm, t_R (major)=9.8 min, t_R (minor)=11.3 min); $[\alpha]_D^{15}$ −221.9 (*c* 1.03, $CHCl_3$) (reported value for 90% ee: $[\alpha]_D^{25}$ −151.5 (*c* 0.51, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.18 (d, *J*=7.8 Hz, 2H, Ar−H), 7.10 (d, *J*=7.8 Hz, 2H, Ar−H), 6.09 (dt, J_1=6.3 Hz, J_2=2.9 Hz, 1H, one proton from CH=C=CH), 5.54 (q, *J*=6.5 Hz, 1H, one proton from CH=C=CH), 2.32 (s, 3H, CH_3), 2.11 (qd, J_1=6.9 Hz, J_2=2.5 Hz, 2H, CH_2), 1.53−1.16 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.3 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=204.8, 136.2, 132.1, 129.2, 126.4, 95.0, 94.3, 31.9, 29.4, 29.3, 29.20, 29.17, 28.8, 22.7, 21.1, 14.1; MS (EI) *m*/*z* (%): 242 (M^+, 1.26), 129 (100); IR (neat): ν=2955, 2924, 2854, 1949, 1513, 1465 cm^{-1}; HRMS calcd for $C_{18}H_{26}$ [M^+]: 242.2035, found: 242.2037.

*4.2.8. (R)-1-(3-Methylphenyl)undeca-1,2-diene (R-**4dj**).* The reaction of $ZnBr_2$ (230.1 mg, 1.0 mmol, 98% from Alfa), **1d** (211.7 mg, 1.5 mmol, 98%), **2j** (240.9 mg, 2.0 mmol), (*S*)-**3** (258.6 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4dj**[6] (105.6 mg, 44%) (eluant: petroleum ether) as a colorless liquid: 92% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 0.7 mL/min, λ=254 nm, t_R (major)=10.3 min, t_R (minor)=11.8 min); $[\alpha]_D^{20}$ −194.9 (*c* 1.02, $CHCl_3$) (reported value for 90% ee: $[\alpha]_D^{25}$ −125.3 (*c* 0.50, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.23−7.14 (m, 1H, Ar−H), 7.14−7.05 (m, 2H, Ar−H), 6.99 (d, *J*=7.2 Hz, 1H, Ar−H), 6.08 (dt, J_1=6.3 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.54 (q, *J*=6.6 Hz, 1H, one proton from CH=C=CH), 2.33 (s, 3H, CH_3), 2.12 (qd, J_1=7.0 Hz, J_2=2.9 Hz, 2H, CH_2), 1.53−1.16 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=205.1, 138.1, 135.0, 128.4, 127.4, 127.2, 123.7, 95.0, 94.5, 31.9, 29.4, 29.3, 29.20, 29.15, 28.8, 22.7, 21.4, 14.1; MS (EI) *m*/*z* (%): 242 (M^+, 1.03), 91 (100); IR (neat): ν=2955, 2924, 2854, 1949, 1606, 1587, 1488, 1464, 1378 cm^{-1}; HRMS calcd for $C_{18}H_{26}$ [M^+]: 242.2035, found: 242.2034.

*4.2.9. (R)-1-(4-Trifluoromethylphenyl)undeca-1,2-diene (R-**4dk**).* The reaction of $ZnBr_2$ (228.7 mg, 1.0 mmol, 98% from Alfa), **1d** (211.9 mg, 1.5 mmol, 98%), **2k** (348.3 mg, 2.0 mmol), (*S*)-**3** (258.4 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4dk**[6] (90.6 mg, 31%) (eluant: petroleum ether) as a colorless liquid: 95% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 0.5 mL/min, λ=254 nm, t_R (minor)=10.4 min, t_R (major)=11.3 min); $[\alpha]_D^{20}$ −151.7 (*c* 1.06, $CHCl_3$) (reported value for 82% ee: $[\alpha]_D^{25}$ −142.5 (*c* 0.55, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.53 (d, *J*=8.1 Hz, 2H, Ar−H), 7.37 (d, *J*=8.4 Hz, 2H, Ar−H), 6.14 (dt, J_1=6.3 Hz, J_2=3.1 Hz, 1H, one proton from CH=C=CH), 5.62 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.14 (qd, J_1=7.1 Hz, J_2=3.0 Hz, 2H, CH_2), 1.54−1.15 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=206.1, 139.1 (d, *J*=1.5 Hz), 128.4 (q, *J*=32.2 Hz), 126.6, 125.4 (q, *J*=3.9 Hz), 124.3 (q, *J*=270.0 Hz), 95.7, 93.8, 31.8, 29.4, 29.3, 29.2, 29.1, 28.5, 22.7, 14.1; ^{19}F NMR (282 MHz, $CDCl_3$) δ=−63.0; MS (EI) *m*/*z* (%): 296 (M^+, 1.15), 129 (100); IR (neat): ν=2927, 2856, 1950, 1616, 1467, 1324, 1164, 1124, 1066, 1017 cm^{-1}; HRMS calcd for $C_{18}H_{23}F_3$ [M^+]: 296.1752, found: 296.1756.

*4.2.10. (R)-1-(3-Methoxyphenyl)undeca-1,2-diene (R-**4dl**).* The reaction of $ZnBr_2$ (230.9 mg, 1.0 mmol, 98% from Alfa), **1d** (212.0 mg, 1.5 mmol, 98%), **2l** (272.2 mg, 2.0 mmol), (*S*)-**3** (258.3 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4dl**[6] (126.6 mg, 49%) (eluant: petroleum ether) as a colorless liquid: 94% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:1, 0.6 mL/min, λ=214 nm, t_R (major)=6.4 min, t_R (minor)=7.3 min); $[\alpha]_D^{19}$ −209.1 (*c* 1.00, $CHCl_3$) (reported value for 94% ee: $[\alpha]_D^{25}$ −193.5 (*c* 0.33, $CHCl_3$)); ^{1}H NMR (400 MHz, $CDCl_3$) δ=7.21 (t, *J*=7.8 Hz, 1H, Ar−H), 6.91−6.83 (m, 2H, Ar−H), 6.74 (dd, J_1=8.0 Hz, J_2=2.4 Hz, 1H, Ar−H), 6.09 (dt, J_1=6.4 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.56 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 3.80 (s, 3H, OCH_3), 2.12 (qd, J_1=7.1 Hz, J_2=2.8 Hz, 2H, CH_2), 1.53−1.16 (m, 12H, $CH_2\times6$), 0.87 (t, *J*=6.8 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$) δ=205.2, 159.8, 136.6, 129.4, 119.2, 112.3, 111.7, 95.2, 94.5, 55.1, 31.8, 29.4, 29.3, 29.19, 29.15, 28.7, 22.7, 14.1; MS (EI) *m*/*z* (%): 258 (M^+, 3.83), 160 (100); IR (neat): ν=2925, 2854, 1950, 1598, 1490, 1466, 1155, 1087, 1047 cm^{-1}; HRMS calcd for $C_{18}H_{26}O$ [M^+]: 258.1984, found: 258.1985.

*4.2.11. (R)-1-Phenyltrideca-1,2-diene (R-**4bb**).* The reaction of $ZnBr_2$ (229.3 mg, 1.0 mmol, 98% from Alfa), **1b** (257.3 mg, 1.5 mmol, 98%), **2b** (212.5 mg, 2.0 mmol), (*S*)-**3** (259.4 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4bb**[4c] (117.9 mg, 46%) (eluant: petroleum ether) as a colorless liquid: 93% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 1.0 mL/min, λ=214 nm, t_R (major)=6.2 min, t_R (minor)=6.8 min); $[\alpha]_D^{26}$ −183.5 (*c* 0.94, $CHCl_3$) (reported value for 95% ee: $[\alpha]_D^{20}$ −181.6 (*c* 1.05, $CHCl_3$)); ^{1}H NMR (300 MHz, $CDCl_3$) δ=7.34−7.26 (m, 4H, Ar−H), 7.22−7.13 (m, 1H, Ar−H), 6.12 (dt, J_1=6.6 Hz, J_2=3.1 Hz, 1H, one proton from CH=C=CH), 5.56 (q, *J*=6.7 Hz, 1H, one proton from CH=C=CH), 2.12 (qd, J_1=7.1 Hz, J_2=3.0 Hz, 2H, CH_2), 1.54−1.15 (m, 16H, $CH_2\times8$), 0.88 (t, *J*=6.6 Hz, 3H, CH_3); ^{13}C NMR (75 MHz, $CDCl_3$) δ=205.1, 135.1, 128.5, 126.5, 95.1, 94.5, 31.9, 29.64, 29.60, 29.4, 29.3, 29.19, 29.17, 28.8, 22.7, 14.1.

*4.2.12. (R)-3-Cyclohexyl-1-phenyl-1,2-propadiene (R-**4eb**).* The reaction of $ZnBr_2$ (229.3 mg, 1.0 mmol, 98% from Alfa), **1e** (165.8 mg, 1.5 mmol, 98%), **2b** (212.6 mg, 2.0 mmol), (*S*)-**3** (259.5 mg, 1.0 mmol,

98%), and toluene (3 mL) afforded (*R*)-**4eb**[4c] (87.8 mg, 44%) (eluant: petroleum ether) as a colorless liquid: 95% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:0, 1.0 mL/min, λ=214 nm, t_R (major)=7.4 min, t_R (minor)=8.3 min); $[\alpha]_D^{25}$ −326.3 (*c* 1.02, $CHCl_3$) (reported value for 99% ee: $[\alpha]_D^{24}$ −335.2 (*c* 1.01, $CHCl_3$)); ^1H NMR (300 MHz, $CDCl_3$) δ=7.34–7.22 (m, 4H, Ar–H), 7.22–7.12 (m, 1H, Ar–H), 6.15 (dd, J_1=6.3 Hz, J_2=3.0 Hz, 1H, one proton from CH=C=CH), 5.56 (t, *J*=6.2 Hz, 1H, one proton from CH=C=CH), 2.21–2.04 (m, 1H, CH from cyclohexyl), 1.92–1.55 (m, 5H, protons from cyclohexyl), 1.39–1.07 (m, 5H, protons from cyclohexyl); ^{13}C NMR (75 MHz, $CDCl_3$) δ=204.1, 135.2, 128.5, 126.6, 126.4, 101.0, 95.4, 37.6, 33.2, 33.1, 26.1, 26.0.

*4.2.13. (R)-((4-Cyclohexylbuta-2,3-dienyloxy)methyl)benzene (R-**4fm**).* The reaction of $ZnBr_2$ (230.3 mg, 1.0 mmol, 98% from Alfa), **1f** (219.6 mg, 1.5 mmol, 98%), **2m** (223.9 mg, 2.0 mmol), (*S*)-**3** (258.7 mg, 1.0 mmol, 98%), and toluene (3 mL) afforded (*R*)-**4fm**[4c] (139.8 mg, 58%) (eluant: petroleum ether/ethyl ether=400:1) as a colorless liquid: 98% ee (HPLC conditions: Chiralcel OD-H column, hexane/*i*-PrOH=100:1, 1.0 mL/min, λ=214 nm, t_R (major)=5.3 min, t_R (minor)=5.7 min); $[\alpha]_D^{27}$ −83.2 (*c* 1.02, $CHCl_3$) (reported value for 98% ee: $[\alpha]_D^{21}$ −82.9 (*c* 0.6, $CHCl_3$)); ^1H NMR (300 MHz, $CDCl_3$) δ=7.39–7.23 (m, 5H, Ar–H), 5.34–5.15 (m, 2H, CH=C=CH), 4.54 (s, 2H, OCH_2Ar), 4.04 (dd, J_1=6.8, J_2=2.3 Hz, 2H, C=C=CCH_2O), 2.10–1.92 (m, 1H, CH from cyclohexyl), 1.86–1.54 (m, 5H, protons from cyclohexyl), 1.39–1.00 (m, 5H, protons from cyclohexyl); ^{13}C NMR (75 MHz, $CDCl_3$) δ=203.9, 138.3, 128.3, 127.8, 127.5, 97.9, 89.3, 71.6, 68.8, 37.0, 33.04, 33.00, 26.1, 25.9.

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Supplementary data

^1H, ^{13}C NMR and HPLC spectra of all products. Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.tet.2013.07.021.

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